

WRIST SPECIAL TESTS

I. CARPAL TUNNEL SYNDROME TESTS

1. PHALEN'S TEST

Purpose

To assess Carpal Tunnel Syndrome (Median nerve compression).

Procedure

- Ask the patient to flex both wrists maximally.
- Dorsum of both hands are pressed together.
- Hold for 60 seconds.

Positive Sign

Pain, tingling or numbness in the thumb, index, middle and lateral half of ring finger.

Indicates

Carpal Tunnel Syndrome.

2. REVERSE PHALEN'S TEST (PRAYER TEST)

Purpose

To assess Carpal Tunnel Syndrome.

Procedure

- Ask the patient to press both palms together.
- Maintain maximum wrist extension for 60 seconds.

Positive Sign

Pain or paresthesia in the median nerve distribution.

Indicates

Carpal Tunnel Syndrome.

3. TINEL'S SIGN (WRIST)

Purpose

To assess Median nerve irritation.

Procedure

- Tap gently over the median nerve at the carpal tunnel.

Positive Sign

Tingling or electric shock sensation in the median nerve distribution.

Indicates

Carpal Tunnel Syndrome.

II. TENDON PATHOLOGY TEST

4. FINKELSTEIN'S TEST

Purpose

To assess De Quervain's Tenosynovitis.

Procedure

- Patient places the thumb inside the fist.
- Fingers wrap around the thumb.
- Examiner passively deviates the wrist towards the ulnar side.

Positive Sign

Pain over the radial styloid.

Indicates

De Quervain's Tenosynovitis.

III. LIGAMENT STABILITY TEST

5. WATSON (SCAPHOID SHIFT) TEST

Purpose

To assess Scapholunate ligament instability.

Procedure

- Apply pressure over the scaphoid tubercle.
- Move the wrist from ulnar deviation to radial deviation.

Positive Sign

Pain or palpable clunk.

Indicates

Scapholunate ligament injury.

IV. THUMB TEST

6. GRIND TEST

Purpose

To assess 1st Carpometacarpal (CMC) joint osteoarthritis.

Procedure

- Apply axial compression to the thumb.
- Simultaneously rotate the thumb metacarpal.

Positive Sign

Pain or crepitus.

Indicates

1st CMC joint osteoarthritis.

I. TESTS FOR BONE, LIGAMENT, CAPSULE & JOINT INSTABILITY

1. ULNAR FOVEA TEST

Purpose

To assess Triangular Fibrocartilage Complex (TFCC) injury.

Procedure

- Press the examiner's thumb into the ulnar fovea (between the ulnar styloid and flexor carpi ulnaris tendon).

Positive Sign

Pain or tenderness over the ulnar fovea.

Indicates

TFCC injury or ulnotriquetral ligament injury.

2. ULNO-MENISCO-TRIQUETRAL DORSAL GLIDE TEST

Purpose

To assess ulnotriquetral ligament and TFCC pathology.

Procedure

- Stabilize the distal ulna.
- Apply a dorsal glide to the triquetrum while assessing pain and movement.

Positive Sign

Pain or excessive movement.

Indicates

Ulnotriquetral ligament injury or TFCC lesion.

3. BUNNELL-LITTLER TEST

Purpose

To differentiate intrinsic muscle tightness from joint capsule tightness.

Procedure

- Flex the PIP joint with the MCP joint extended.
- Repeat with the MCP joint flexed.

Positive Sign

- PIP flexion improves when MCP is flexed → Intrinsic muscle tightness.
- PIP flexion remains limited → Capsular tightness.

Indicates

Intrinsic muscle contracture or capsular restriction.

★ REVISION TABLE – WRIST & HAND SPECIAL TESTS

Category	Test	Positive Sign	Indicates
Carpal Tunnel	Phalen's Test	Tingling/numbness	Carpal Tunnel Syndrome
	Reverse Phalen's Test	Tingling/numbness	Carpal Tunnel Syndrome
	Tinel's Sign	Tingling over median nerve	Carpal Tunnel Syndrome
Tendon Pathology	Finkelstein's Test	Pain over radial styloid	De Quervain's Tenosynovitis
Ligament Injury	Watson Test	Pain/Clunk	Scapholunate ligament injury
Thumb Pathology	Grind Test	Pain/Crepitus	1st CMC Osteoarthritis

SPECIAL TESTS

I. TESTS FOR CIRCULATION & SWELLING

1. ALLEN TEST

Purpose

To assess the patency of the radial and ulnar arteries.

Procedure

- Patient clenches the fist.
- Compress both radial and ulnar arteries.
- Patient opens the hand.
- Release one artery at a time and observe colour return.

Positive Sign

Delayed or absent return of normal hand colour.

Indicates

Arterial insufficiency or vascular occlusion.

2. DIGITAL BLOOD FLOW TEST

Purpose

To assess circulation to the fingers.

Procedure

- Compress the fingertip until it blanches.
- Release the pressure.

Positive Sign

Delayed capillary refill (>2 seconds).

Indicates

Reduced peripheral circulation.

3. FIGURE-OF-8 MEASUREMENT

Purpose

To assess hand and wrist swelling (edema).

Procedure

- Use a measuring tape in a standardized figure-of-8 pattern around the wrist and hand.
- Record the measurement.

Positive Finding

Increased measurement compared with the opposite side.

Indicates

Hand or wrist edema.

4. HAND VOLUME TEST**Purpose**

To assess hand edema.

Procedure

- Immerse the hand into a volumeter filled with water.
- Measure the volume of displaced water.

Positive Finding

Increased displaced volume.

Indicates

Hand swelling/edema.

II. TESTS FOR THORACIC OUTLET SYNDROME**1. ADSON'S TEST****Purpose**

To assess Thoracic Outlet Syndrome (TOS).

Procedure

- Palpate the radial pulse.
- Patient extends the neck, rotates the head toward the tested side, and takes a deep breath.

Positive Sign

Reduced/absent radial pulse or reproduction of symptoms.

Indicates

Thoracic Outlet Syndrome.

2. ROOS STRESS TEST (EAST)**Purpose**

To assess Thoracic Outlet Syndrome.

Procedure

- Patient abducts shoulders to 90°, externally rotates them, and flexes elbows to 90°.

- Patient repeatedly opens and closes the hands for 3 minutes.

Positive Sign

Pain, numbness, tingling, heaviness, or inability to continue.

Indicates

Thoracic Outlet Syndrome.

★ REVISION TABLE – WRIST & HAND SPECIAL TESTS

Category	Test	Indicates
Bone/Ligament	Ulnar Fovea Test	TFCC injury
	Ulna-Menisco-Triquetral Dorsal Glide	TFCC/Ulnotriquetral ligament injury
	Watson Test	Scapholunate ligament injury
	Bunnell-Littler Test	Intrinsic muscle or capsular tightness
Circulation & Swelling	Allen Test	Radial/Ulnar artery patency
	Digital Blood Flow	Peripheral circulation
	Figure-of-8 Measurement	Hand edema
	Hand Volume Test	Hand edema
Thoracic Outlet Syndrome	Adson's Test	Thoracic Outlet Syndrome
	Roos Stress Test	Thoracic Outlet Syndrome